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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025**25VDE102 – EMBEDDED SYSTEM DESIGN**

(For VDE)

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Module**.**Module – I**

	Marks	BT*	CO*	PO*
1. a) What is RTL? Explain RTN with instruction set architectural expressed in RTN.	10	2	1	3
b) Elucidate Rise and Fall Times and Propagation Delay.	10	2	1	3
2. a) Summarize the concept of Structural Faults.	10	2	1	3
b) Explain a First-Order Resistor Model with the effects of parallel components.	10	2	1	3

Module – II

3. a) What is Unified Modeling Language? Why is it used? Explain use case with the use case description.	10	2	2	3
b) What is Dynamic Modelling with Unified Modelling Language? What are the elements of Dynamic Modelling?	6	2	2	3
c) Explain Create and Destroy.	4	2	2	3
4. a) Illustrate				
i) Fork and Join.	10	2	2	3
ii) Branch and Merge.	10	2	2	3
b) Explain Data and Control Flow Diagrams.				

Module – III

5. a) What are life cycle models? What are the four life-cycle models? Summarize Rapid Prototyping model.	10	2	3	3
b) How do you quantify the system? Explain.	10	2	3	3
6. a) Illustrate the six steps to Design an embedded system with example.	6	2	3	3
b) What is the goal of architectural design? Explain the mapping of function to hardware.	10	2	3	3
c) What is the goal of requirement identification process.	4	2	3	3

Module – IV

7. a) Explicate Black Box, Grey Box and White Box Tests with examples.	10	2	4	3
b) Explain Bridge Fault with an example.	10	2	4	3
8. a) Explain Path Sensitizing with example for Single Variable Single Path.	10	2	4	3
b) Explain Scan Design Testing.	10	2	4	3

Module – V

9. a) What is an Operating System? What is the Kernel? Explain any three services of the Kernel.	10	2	5	3
b) Explain 'Task' in 'Vxworks' context. Explain the different possible states of a Task under Vxkernel.	10	2	5	3
10. a) Explain the Kernel Services of Vxworks.	10	2	5	3
b) Explain the Watchdog timer operation under VxWorks Kernel.	10	2	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
